Nuthanan Tharmarajah

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Education

University of Waterloo — Waterloo, ON
Bachelor of Software Engineering — *Aug. 2024 – May 2029*

- **GPA:** 3.9/4.0
- **Activities and Societies**: UW Blueprint, UW Formula Electric, SE Yearbook Representative, Tamil Student Association

Technical Skills

Languages: Swift, Go, Python, Java, TypeScript, JavaScript, Ruby, C/C++
Frameworks: React, Next.js, FastAPI, Node.js, Flask, Express, TailwindCSS, LangChain
Systems/Infra: Redis, Apache Airflow, Docker, Kubernetes, GitHub Actions, PostgreSQL, MongoDB, Supabase, RunPod
Cloud/Tools: AWS (EC2, Lambda, CloudFront), GCP (APIs, OCR, Storage), OpenShift, Elastic, Selenium, 1Password, Git

Experience

Software Development Engineer — Jan. 2026 – April 2026
Royal Bank of Canada (RBC) – Return Offer Received — *Toronto, ON*

- Built **LangGraph-based** AI tooling using Next.js, Flask, and MongoDB to evaluate JIRA requirements and generate test cases, automating workflows and improving documentation for **160+ developers, PMs, and QEs**
- Co-developed **SSL/TLS renewal automation** by abstracting certificate generation from Bash into APIs, integrating MCP-based tooling, and scripting provisioning requests with Playwright across **4 internal services**
- Developed an **AI-powered data compliance platform** using Next.js, MongoDB, and LangChain, replacing email threads with an LLM-validated form for production data movement, reducing turnaround from **5 days to 5 minutes**
- Optimized a **legacy 8,100+ line Java BDD Selenium framework**, reducing regression test execution time by **50%**

Software Development Engineer — May 2025 – August 2025
Guelph Eye MD – Return Offer Received — *Guelph, ON*

- Built a **Raspberry Pi-based health card scanner** that decrypts patient data and triggers an Electron interface via StreamDeck for one-click patient check-in, reducing overall staff check-in and reception time by **30%**
- Designed a **consent form processing pipeline** using Golang and Electron, integrating GCP OCR to parse fields and structure scanned forms into Google Sheets, reducing manual processing by **90%** across **120+** weekly faxes
- Owned the design of **PostgreSQL schemas and data models** for **300+ weekly patient events**, and built a daily scheduled **Apache Airflow ETL pipeline** to ingest and normalize clinic data for reliable analytics
- Refactored backend services and CI/CD pipelines, implemented centralized **secrets management** with 1Password and enforced **HIPAA-compliant** workflows, strengthening production-grade reliability and security posture

Lead Software Developer — May 2025 – Present
SSICSIM 2026 — *Toronto, ON*

- Built a conference management platform **(Next.js, FastAPI, PostgreSQL)** for **50+ staff** and **380+ delegates**
- Defined database architecture with **20+ REST API endpoints** and **18 schemas** for delegates, committees, and roles
- Engineered **Redis-backed job queues** automating **140+ weekly emails**, and deployed Docker-containerized services on Render with Supabase PostgreSQL via CI/CD pipelines (GitHub Actions) and agentic validation

Projects

Roomtastic | *Next.js, FastAPI, PostgreSQL, Three.js, Python* — April 2026

- Built a **3D interior design platform** using **Three.js/WebGL** where users design rooms with **AI-generated furniture meshes** from a built-in catalogue or vendor URL, generating a **direct shopping list** for one-click purchasing
- Deployed and optimized an **open-source 2D image to 3D generation model** on RunPod GPU infrastructure, reducing inference cost by **92% ($0.50 → $0.04 per generation)** by self-hosting instead of external APIs

Solana Swift SDK | *Swift, Solana JSON-RPC, Web3* — December 2025
Funded by Solana Foundation ($40,000 USD grant)

- Built a **3,800+ line Swift SDK** for Solana, enabling iOS apps to integrate **3+ wallets** (Phantom, Solflare, Backpack)
- Abstracted **Web3Core and RPC layers** into clean interfaces for session management, signing, and transaction execution
- Designed a **type-safe transaction builder** with JSON-RPC, supporting System, Token, Associated Token, and Memo programs, mirroring established Solana SDK interfaces

Solshare | *Swift, Firebase, Cohere API, Solana, Typescript* — September 2025
Winner at Hack the North 2025 — Best Use of Cohere API & Best Use of Solana

- Built an **iOS bill-splitting app** in Swift for receipt tracking, item assignment, and dispute resolution among friends, with **Firebase WebSocket** for real-time sync for instant expense, balance, and assignment updates
- Developed a **self-verifying LLM pipeline (Cohere Command + Vision)** using reasoning models to automatically itemize and extract receipt data, eliminating manual expense entry